SolarLab vs SCAPS-1D — Parity Campaign Status & Verdict

2026-06-23

Bottom line

On the validated `scaps_mirror_v2` configuration, SolarLab reproduces SCAPS-1D's **physics behaviour** — judged on the trends-over-absolutes standard (sweep *direction* and *magnitude fidelity*, with base absolutes required only to *approach* SCAPS):

- **Base operating point — matched by physics.** Effective-DOS band potentials (`dos_band_potentials`) + heterointerface Auger de-spike (`het_recomb_despike`) land V_{oc} on SCAPS within a few mV — *no fit knobs*.
- **All 11 single-variable sweep DIRECTIONS — matched by physics.** The steady-state interface-plane-states driver recovers every sweep's trend direction, including the ETL-donor-doping rising arm that the transient path gets backwards.
- **Sweep MAGNITUDES — structural deficit (documented boundary).** The response *amplitudes* (e.g. Nd_ETL ~52 %, PVK/ETL N_t ~63 % of the SCAPS V_{oc} range) are not fully reproduced. This is not a tuning gap — see Section 3.

Verdict: the model is validated for SCAPS-comparison use. Remaining gaps are documented engine boundaries, honestly reported, not fudged.

1. Base operating point (no fit knobs)

Metric	SolarLab (SS iface-states)	SolarLab (transient f=0.53)	SCAPS	diff
V_{oc} (V)	1.167	1.164	1.168	−0.1 … −4 mV
J_{sc} (mA/cm^2)	25.72	25.72	26.28	−2.1 % (front-reflection residual)
FF (%)	88.8	87.3	87.0	+0.3 … +1.8
PCE (%)	26.5	26.1	26.7	−0.2 … −0.6

2. Sweep directions (11/11 by physics)

All eleven single-variable sweeps (CBO, ETL/PVK doping, the four interface/bulk N_t, the four E_t) match the SCAPS trend direction under the steady-state interface-plane-states driver. Detailed four-config overlay (SCAPS / transient f=0.53 / f=0.66 / SS interface-states), all four figures of merit per sweep: `SolarLab_SCAPS_validation_2026-06-22.pdf`.

3. Documented boundaries (not matched — and why)

1. **Sweep magnitudes (structural).** Root cause, established by a 10-agent adversarial investigation: SolarLab's V_{oc} tracks the doping-swung built-in potential (the V_{bi} Poisson boundary condition swings

+59.5 mV/decade with ETL doping), whereas SCAPS models **fixed-work-function flat-band contacts** whose V_{oc} responds only with the gentle $kT/q{\cdot}\ln(N_D)$ shift; this is compounded by a bulk-Auger-limited, doping-insensitive V_{oc} floor. The per-interface rate calibration is a *rate knob* (it sets the base/absolutes, not the sweep amplitude), so it cannot close this. Reaching it requires a SCAPS-faithful **quasi-Fermi-variable contact engine** — a large research rewrite. The flat-band-contact convention does not converge in the present steady-state driver (it needs that engine), so the magnitude deficit stands as a boundary.

2. **Deep conduction-band-offset cliff ($\Delta E_C \leq -0.2$ eV).** The collapsed-junction steady state is unreachable by *any* algebraic Newton (coupled Newton, decoupled Gummel, Anderson acceleration, and pseudo-transient continuation all stall 10–15 orders above tolerance). It is computed instead by a certified transient settle; since the conduction-band offset (not interface recombination) sets V_{oc} there, the steady state equals the transient, so this regime carries the transient value (shown as hollow markers in the CBO figure).
3. **+0.5 eV spike-side J_{sc} collapse.** Mechanism understood (the thermionic collection knee), not fully reproduced. A narrow extreme, low practical impact.

4. Canonical SCAPS-comparison configuration

`configs/scaps_mirror_v2.yaml` with `dos_band_potentials` (default on), `het_recomb_despike: 0.53`, and the steady-state driver run with `iface_states=True`. The transient (MOL) solver remains available for the dynamics SCAPS cannot model.

5. A note on the standalone SCAPS engine (abandoned)

A standalone "SCAPS-faithful" engine subpackage was prototyped (2026-06-02) and abandoned (2026-06-03): it rode the existing MOL transient (fast scan, ions off = quasi-steady-state) plus two reference-fit calibration knobs, and could not reach SCAPS absolutes *by physics* (only by tuning). Its diagnostics became the de-spike (base) and interface-plane-states (directions) work now on `main`; the standalone engine was not retained.

6. Recommendation

SCAPS parity has reached its physics-bounded plateau — directions and base are matched by physics; absolute magnitudes need a large rewrite with uncertain payoff and are out of scope under the trends-over-absolutes standard. SolarLab's distinctive scientific value is the physics SCAPS *cannot* model — transient dynamics (hysteresis, impedance, degradation, transient photovoltage) and 2D microstructure / tandem / design-space studies. Recommend pivoting there.